Subhash Sarangi

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SUMMARY

Full Stack Software Engineer with 4+ years of experience building scalable web applications using React, Typescript, and SpringBoot. Experienced in developing end-to-end features including frontend interfaces, backend APIs, and performance optimization. Strong in system design, reusable component libraries, and test automation.

CORE SKILLS

Languages: Java, Typescript, Javascript, Python, C#, Bash
Frontend: React, Redux, Lit, Playwright
Backend: SpringBoot, Dotnet, REST APIs, Microservices
Databases & Tools: PostgreSQL, Redis, Docker, Git, Kubernetes (Overview)
Platforms: Linux, AWS
AI: RAG, GraphRAG, MCP

EXPERIENCE

Software Engineer Aug 2022 – Present
GE Vernova, Bangalore, India

- **Backend System Integration (SharePoint)**
  - Designed and implemented a backend integration between Microsoft SharePoint and internal document services using SpringBoot
  - Enabled secure and controlled document access for end users
  - Improved backend performance and scalability using Redis-based caching, reducing redundant external API calls and latency
- **Micro-Frontend Development (Asset 360)**
  - Built a React-based micro-frontend for asset insights with configurable dashboards
  - Designed and implemented RESTful APIs using SpringBoot and PostgreSQL for dynamic data retrieval and configuration
  - Setup the complete CI/CD pipeline using Jenkins
  - Delivered end-to-end features integrating frontend UI with backend services
- **Design System (DEX)**
  - Co-developed a modular design system with 50+ reusable UI components using Lit
  - Accelerated frontend development across multiple applications
  - Implemented component-level automation testing, reducing integration bugs by 40%
- **Testing & Engineering Contributions**
  - Automated 60+ end-to-end test cases using Playwright, reducing regression effort by 85%
  - Built workflow to auto-generate manual test cases, reducing documentation effort by 60%
  - Developed a lightweight state management library (Tiny Store) with bundle size under 1 KB
  - Built a reusable HTML-to-PDF library adopted across multiple teams
  - Resolved 50+ critical customer issues, actively collaborating with cross-functional business and product teams ensuring a mean time to resolution under 30 days

Projects

Wiki-base

- Built a self-hosted RAG system in **Python** for querying document collections with grounded citations
- Implemented document chunking, vector embeddings, **PostgreSQL/pgvector** storage and configurable LLM providers
- Designed GraphRAG retrieval pipeline using entity extraction, OpenIE-based knowledge graph constructtion and asynchronous background workers.
- Added support for multiple retrieval strategies including vector search, HippoRAG-based ranking and fact-oriented knowledge graph traversal.
- Built REST APIs for document ingestion, retrieval and conversational querying.

EDUCATION

Integrated MTech in Computer Science 2017 – 2022
University of Hyderabad
CGPA: 8.23/10
Thesis: Understanding Visuo-Textual Reasoning

ACHIEVEMENTS

- Impact Award, GE (Engineering Recognition Day 2023)
- Winner – Hackmakers BuildWithAI Hackathon (Rank 1 among 4500 participants)
- Codechef Certified Programmer (CCDSAP)
- AIR 10 – Code-In-Lockdown (IEEE)
- Semi-finalist – HCL CodeForCovid Hackathon (Top 32 out of 7500 teams)